INSY 662 Final

Emery Dittmer -260658030

1.1. Describe all data pre-processing tasks that should be performed before separating the data into training and test/validation dataset.

First, look at missing or NA records, and if there are few of them, we should drop or somehow mark them. (less than 10% of the data)

Next, we will want to identify and discard identifier fields. Unique ID fields help make sense of the data but confuse the algorithms. In this case, *Tail_Number.*

Next, we may want to investigate and drop outliers from our dataset, as these might be errors.

Next, we may want to dummify categorical data.

Next, we can sort through the fields and remove redundant ones.

Lastly, we split the data based on a split ratio, and a random_state.

1.2. You recognize that there are about 100 records where the value of *distance_between_airport* is missing. What should we do with them?

Either:
1. Drop these observations from the dataset.
2. Fill these values in with the average distance from the dataset. Since all distance observations are for the same trip.
3. Replace missing distances based on day and weather. The route may vary based on these.

1.3. Which variable(s) should be excluded from the analysis? Why? (words)

Tail_Number is a unique Identifier which does not add value to the analysis and should be excluded.
Day_of_Month, and *Day_of_Week* should be removed as they are redundant with *Flight_Date*.
Origin_Airport and *Destination_Airport* will all be YUL and JFK respectively, so they can be excluded.

1.4. In your opinion, what would be the most appropriate technique/algorithm for this predictive task? Why? (words)

A decision tree technique would help best classify the lateness factor based on all the variables. Decision tree models are methods optimized for using categorical and numerical data to group data based on predictions. The easy interpretability makes decision trees an ideal technique for this task.

1.5. After running the algorithm of your choice, using all predictors excluding those from 3), you find that the performance of your model is horrible. What can be done to improve the predictive performance? (words)

A primary component analysis will detect variables that are not contributing to the predicting power of the model so that we can remove them. I might also try tuning model hyperparameters. If those do not work, a different model like a random forest, boosted forest, or Artificial Neutral Network may compensate for bias more easily.

2.1. Among algorithms that are covered in the course (you may include those in Team Project 2 presentations), list algorithms that are black-box and algorithms that are transparent.

Black box	Transparent
Artificial Neural Network (ANN)	Linear Regression
Random Forest	Ridge
Gradient Boosting	Decision Tree
Convoluted Neural Network (CNN)	Logistic regression
K Nearest Neighbors (KNN)	
Multivariate Adaptive Regression Splines	
Extra Tree	
Adaboost	

2.2. For each algorithm that is black-box, briefly explain why the algorithm is black-box. For each algorithm that is transparent, briefly explain how we can interpret its results.

ANNs build networks of nodes with specific weights but are complicated to interpret the weight meaning. Similarly, Random forests, boosting, CNNs and others do not provide insights on how weights are assigned or how they approximate the underlying function. KNN, meanwhile, seeks to group all features without any explicit prediction. These clusters can be very difficult to interpret.

We understand the decision-making process with Decision trees, and the impact of each feature on a prediction. Similarly, linear regression, ridge and logistic regression assign coefficients to each predictor that demonstrate the link between features and their impact on the final prediction.

2.3. Random Forest algorithm and Classification and Regression Tree (CART) algorithm are highly relevant to each other. Explain why one algorithm is black-box while another is transparent.

CART and random forests both rely on splitting out data based on binary splits. CART is a transparent algorithm that evaluates all possible data splits and recursively partitions them into subsets with similar target values before picking the optimal path. We understand CART's decision-making and can evaluate the predictor's influence easily.

Random forests rely on a strategy of iterative decision trees with bagging, where observations are randomly grouped for a decision tree. Many trees make up a forest. The forest uses the combined decision-making from each tree to form a final model. However, the relationship between predictors is not clear, nor is the model methodology.

2.4. What can we do if we are required to use a black-box algorithm, but we are also required to interpret the results?

We can use algorithms that allow for greater explainability than others. Comparing the number of observations in a cluster and their centers in KNN leads to understanding cluster characteristics. Random forests and boosting algorithms offer methods for constructing dendrograms for the best fit model which will help with interpretability.

The best interpretations for CNN models come from predictions. Having a test dataset to make predictions may help with the interpretability of the model. For instance, predicting observations that differ slightly in one predictor and noting their results.

3.1. Define the problem. What exactly are you predicting/classifying?

We are predicting whether a user/group of users will watch an ad for the defined period on the online video platform.

3.2. What data would you need? (i.e., variables, data type, etc.) How are you going to obtain those data? (i.e., data source, collecting method, etc.)

We need to know the user's basic information, watch history, and skip interactions. The ad details (category, language) and details about the video being watched after the ad will help.

Assuming there is a user login system as some videos are age-restricted. Internal databases will contain user biographic and watch history data. The video details are likely internally stored as well, as uploaders classify their videos to enable internal recommendation features. If ads are on this site, ad data should be internal and readily available.

Services exist to build databases on videos and advertisements if data is missing.

3.3. What are potential constraints or obstacles that you might anticipate when your proposed model is used in practice?

The data may be noisy, leading to poor results. Ads for users while travelling may need to be clarified for the model as some may be in foreign languages.

Insufficient predictors may cause false predictions. The model may not capture the user's motivations or cultural background. The target video may be viral, so people are unwilling to wait longer than they must for ads.

While the model may predict ad watch time and revenue from ads, you must decide what to do with the data. This is a practical consideration for the implementation of model insights. You will need to balance revenue and alienate your consumers who may substitute your site for another.

4.1. What is the main conclusion of this study?

The study concludes that linear multiple regression models are an accurate method for predicting the number of daily active COVID cases for the next day in India.

4.2. Do you agree with the conclusion? Justify your answer (e.g., if you agree, why? If you do not agree, why not?)

I disagree with the conclusion.

The data on the number of cases suggests an exponential relationship, not linear as the paper uses. A logistic or polynomial model comparison is needed.

Linear relationships assume a continual increase with x (time). At some point, the model will predict more new cases for the next day than people in India. Introducing splines or discontinuity will become necessary when cases decrease.

The coefficient's p-values in table 6 are above 0.05, meaning they are not statistically significant predictors. It is possible that the linear combination contributes to better p-values, but the paper fails to mention it.

Active cases should follow some form of equation 1, however, the model suggests equation 2. A deceased person should not also contribute to active cases.

$$\sum Active\ Cases_{t} = \beta_0 + \beta_1 \sum_{t=0}^{t-1} Active\ Cases + new\ cases - \beta_2 \sum Recovered - \beta_3 \sum Deceased$$

Equation 1. Expected Active cases Math

$$\sum Active\ Cases_{t} = \beta_0 + \beta_1 \sum_{t=0}^{t-1} Active\ Cases - \beta_2 \sum Recovered + \beta_3 \sum Deceased$$

Equation 2. Model proposed Active Cases

Other issues arise from the methodology, conclusions and strategies used.

4.3. If you were one of the authors, would you develop a predictive model differently? Why or why not?

I would not change model development if the motivation were to add resources for medical personnel. Linear regressions are quick and easy to visualize, thus demonstrating tangible predictions to officials leading to decisions to increase resources to save lives.

Without this specific motivation, I would develop the model differently. Informing officials is essential, however, 2 or 3 weeks of predictions help adaptation. Modelling on more 'completed' outbreaks like MERS is more helpful for determining life cycle, modelling and mathematical frameworks (linear vs. logistic vs. decision tree) than an incomplete outbreak lifecycle. Lastly, I would try logistic regression, to choose the best model.

4.4. If you were one of the authors and the primary goal was to predict the number of active cases the next day, what would you do to improve the quality of the study in terms of the data? Be specific in your answer as much as possible. Also consider the feasibility of your idea as well as the novelty.

I would get additional data from past outbreaks like MERS (Original data published in 2019). Contextual data would help determine factors, scale the data and outbreak duration.

Adding running counts will help with modelling techniques. Neither running total or linear regression should ever predict a decrease in the number of cases, which validates the use of our linear regression on this type of data.

I would eliminate observations with less than 100 active cases. These values are outliers that throw off the model with irrelevant data. 100 is a threshold that reduces 'useless' observations.

Lastly, I would standardize the data using a combination of min-max and logarithmic scalers. Min-max scaling would scale data to enable comparisons to outbreak datasets (MERS), and give predictions on %increases which may be helpful in some cases. Scaling logarithmically enables linear regressions to model exponential data. The original data (below) supports the argument for logarithmically as data for the same period as in the paper shows a linear relationship in time as opposed to exponential thus making linear modeling easier.

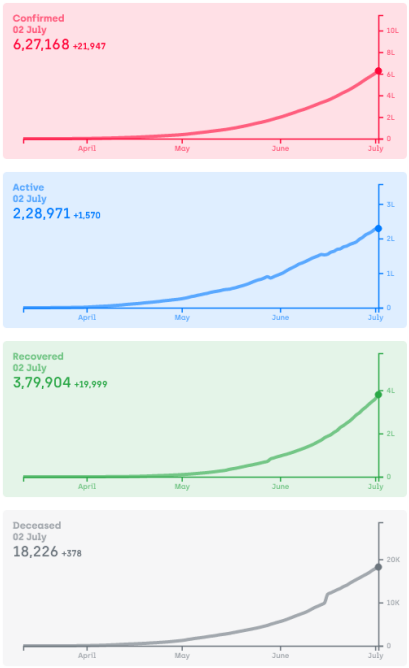


Figure 1: Normal Scaling of Data

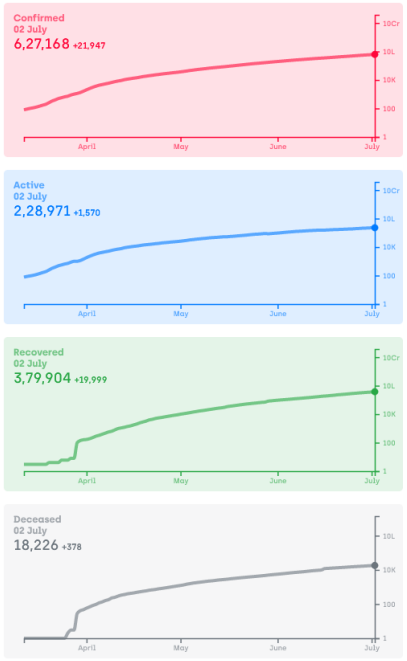


Figure 2: Logarithmic scaling of data